Taehoon Kim

jerry38250325@gmail.com | Google Scholar | Personal Website

Research Interests

My research aims to develop robust perception for real-world, safety-critical environments through multimodal representation learning with computer vision and vision–language models. I am particularly interested in spatial recognition and autonomous-driving perception via multi-sensor fusion (RGB, LiDAR, Radar). I also study industrial and cross-modal anomaly detection via distillation-based teacher–student frameworks.

EDUCATION

Kyonggi University — Sep. 2025 – Present
M.S in Computer Science — *Suwon, Korea*
· Advisor: Junho Ahn
· GPA: 4.5 / 4.5

Kyonggi University — Mar. 2019 – Aug. 2025
B.S in Department of Restaurant and Culinary — *Seoul, Korea*
B.S in Department of Artificial Intelligence — *Suwon, Korea*
· Overall GPA: 4.3 / 4.5 – Summa Cum Laude (Valedictorian)
· AI GPA: 4.3 / 4.5 (42 credits completed)

Experience

K-SW Square, Purdue University — Oct 2023 – Dec 2023
Research Intern — *West Lafayette, IN, USA*
advisor: Eric T. Matson & Anthony Smith

AI Vision Lab, Kyonggi University — May 2024 – Aug 2025
Undergraduate Research Assistant — *Suwon, Korean*
Advisor: Prof. Junho Ahn

National Center of Excellence in Software, Kyonggi University — Sep. 2025 – Present
Graduate Teaching Assistant — *Suwon, Korean*

Military-Academia Joint AI Education Program, ROK DCC — 2024 – 2025
Teaching Assistant — *Suwon, Korean*

Publications

International Journal

[1] **Taehoon Kim**, Sehun Lee, Namgi Kim, Youjin Seo, “XSANet: Industrial X-ray Anomaly Detection via Spatial Transformer and Attention-Enhanced Teacher-Student Framework,” Image and Vision Computing (IMAVIS), Elsevier, *under review*

[2] **Taehoon Kim**, Sehun Lee, Junho Ahn, “Zero-Shot Vision-Based Robust 3D Map Reconstruction and Obstacle Detection in Geometry-Deficient Room-Scale Environments,” Computers, Materials & Continua, vol. 86, no. 2, pp. 1–30, Dec. 2025.

[3] Sehun Lee, **Taehoon Kim**, Sookyun Kim, Junho Ahn, Namgi Kim, “Unsupervised Structural Defect Classification via Real-Time and Noise-Robust Method in Smartphone Small Modules,” Electronics (MDPI), vol. 14, no. 17, art. no. 3455, Aug. 2025.

[4] Sehun Lee, **Taehoon Kim**, Junho Ahn, “Zero-Shot Based Spatial AI Algorithm for Up-to-Date 3D Vision Map Generations in Highly Complex Indoor Environments,” Computers, Materials & Continua, vol. 84, no. 2, pp. 3623–3648, Jul. 2025.

International Conference

[1] Doyoung Koh, **Taehoon Kim**, Sehun Lee, Namgi Kim, “Weighted Knowledge Distillation with DisturbLabel for Robust Learning,” 2026 17th International Conference on Ubiquitous and Future Networks (ICUFN), Milan, Italy, Jul. 2026.

Domestic Journal

[1] **Taehoon Kim**, Jiyoung Na, Jiwon Yoon, Sehun Lee, Junho Ahn, “Multiple Camera-Based Real-Time Long Queue Vision

Algorithm for Public Safety and Efficiency," Journal of the Korea Society of Computer and Information (JKSCI), vol. 29, no. 10, pp. 47–57, Oct. 2024. DOI: 10.9708/jksci.2024.29.10.047

Domestic Conference

[1] **Taehoon Kim**, Sehun Lee, Junho Ahn, "Industrial X-ray UniNet: A Robust Spatial Transformation Architecture for X-ray Anomaly Detection in Smartphone Actuator Modules," Proceedings of the 2025 KAD Winter Conference (Outstanding Paper Award)

[2] Sehun Lee, **Taehoon Kim**, Junho Ahn, "Genetic Neural Networks: Efficient Training Initialization Strategy via Weight Recombination," Proceedings of the 2025 KAD Winter Conference (Best Paper Award)

[3] **Taehoon Kim**, Sehun Lee, SeungBeom Lee, Junho Ahn, "0-VISTA: Zero-Shot Spatial AI for 3D Mapping and Detection in Complex Indoor Environments," Proceedings of the 2025 KAD Spring Conference

[4] Sehun Lee, **Taehoon Kim**, Doyeon Kim, Junho Ahn, "Latent Hotspot Crossover for Disentangled and Interpretable Representation Learning in Variational Autoencoders," Proceedings of the 2025 KAD Spring Conference (Outstanding Paper Award)

[5] Doyeon Kim, SeungBeom Lee, **Taehoon Kim**, Junho Ahn, "Detection of Movable Object Layout Changes in Shared-Use Facilities," Proceedings of the 2025 KAD Spring Conference

[6] **Taehoon Kim**, Jiyong Na, Sehun Lee, Jiwon Yoon, Junho Ahn, "Deep Learning Vision-Based Line Recognition Algorithm Suggestion for Real-Time Intercity Bus Passenger Queue Efficiency," Proceedings of the 2024 KSII Spring Conference

[7] Yechan Lee, **Taehoon Kim**, Jiyoung Na, Jiwon Yoon, Sehun Lee, Hanyong Lee, "Analysis of Autonomous Driving Algorithms for Road Driving of Delivery Robots," Proceedings of the 2023 KIIT Summer Conference

Patent

Deep Learning-Based Queue Recognition System — Oct. 2024
Korea Patent Application — *No. 10-2024-0144283*

Projects

sLLM-Based Query Recommendation System for Efficient Cybersecurity Monitoring in Physically Isolated Networks — Apr. 2026 – Present
ITSA & ROK DCC

Precise 3D Map Reconstruction and Multi-Trajectory Tracking of Evacuees through Zero-Shot Vision–Based Drones in Indoor Disaster Environments — Mar. 2026 – Present
NRF

Pedestrian Safety System for Large-Scale Buildings Using 3D Congestion Mapping with Collaborative Robots and Vision AI for Disaster Response — Sep. 2025 – Aug. 2026
NRF

Acoustic-Based Real-Time Drone Localization & Tracking AI System — Jul. 2025 – Dec. 2025
ITSA & ROK DCC

Improvement of Detect Detection and Classification Capabilities in SUB2 ASS'Y GAP Inspection — Nov. 2024 – Feb. 2025
im co., ltd.

Real-Time Multi-Modal Vision-Integrated Military Fitness Assessment — Sep. 2024 – Dec. 2024
ITSA & ROK DCC

Data Processing for Textile Knowledge Generative AI Development and Utilization — Jul. 2024 – Sep 2024
KTDI

Improvement of Detect Detection and Classification Capabilities in X-RAY Ball Inspection — Jul. 2024 – Aug. 2024
im co., ltd

Enhancing Culinary Experience: Real-Time Interactive Cooking Assistance Utilizing Mixed Reality — Oct. 2023 – Dec. 2023
Purdue University

Awards

Best Paper Award, 2025 Korea Data Science Winter Conference — Dec. 2025

Outstanding Paper Award, 2025 Korea Data Science Winter Conference — Dec. 2025

Outstanding Paper Award, 2025 Korea Data Science Spring Conference — Jun. 2025

3rd Prize, Kyonggi University SW Imagination Company Project Competition — Nov. 2024

1st Prize, Kyonggi University Advanced Capstone Design Project Competition — Jun. 2024

2nd Prize, Kyonggi University Foundational Capstone Design Project Competition — Jun. 2023

1st Prize, Kyonggi University GamsungSW IT Competency Competition — Dec. 2019

Honors

Letter of Appreciation, ROK Defense Communications Command — Apr. 2026
· *Award for outstanding contribution in AI development and mentoring*

AI & SW Talent Scholarship, Noh Youngbaek Scholarship Foundation — Nov. 2025
· *Award for excellence in the AI & Software field*

Faculty Council Honors Scholarship, Kyonggi University — Jun. 2025
· *Award for outstanding academic achievement*

Highest Honors, Kyonggi University — 2025
· *Graduate 1st in Dept. of Restaurant & Culinary*

Semester High Honors, Kyonggi University — 2024
· *Award for perfect GPA of 4.5 / 4.5*

Semester High Honors, Kyonggi University — 2019, 2022 · 2023 (1st Sem), 2024
· *Award for Academic Excellence (1st)*

Technical Skills

Programming languages
Python, C, Java, R

Deep Learning Framework
Pytorch, Tensorflow, Keras

Language Skills

TOEIC – Total Score 920 — Aug. 2024

TOEIC Writing – Advanced Mid (AM) — Apr. 2024

TOEIC Speaking – Intermediate High (IH) — Apr. 2024

English Presentation — Jul. 2024
Kyonggi University National Center of Excellence in Software
· *Presented program highlights and achievements to a CHED delegation, Philippine President's office* — news article

English Translation — Oct. 2023
Purdue Military Research Institution (PMRI) & IITP
· *Translate AI lectures and supported communication for ROK ROTC officers*